

FML Midterm Cheat Sheet

The rule behind every practice question, one topic per page

Data



Labels



Metrics



Models



Trees

Five topics, in the order of the practice set

01	————	Foundations and data	learning types · overfitting · bars · biases Units 1–2 · Q1–11 · four pages
02	————	Labels and linear models	triple barrier · OLS · Lasso · Ridge Units 3–4 · Q12–19 · four pages
03	————	Precision, recall and F1	confusion matrix · rare targets · F1 Q20–31 · four pages
04	————	Classifiers and anomalies	logistic · SVM · LOF · One-Class SVM Units 5–6 · Q32–36 · four pages
05	————	Trees and ensembles	Gini/entropy · bagging · boosting · iForest Unit 7 · Q37–48 · seven pages

Last two pages: the ten triggers worth memorising

01

Foundations and Financial Data

Learning types, overfitting, honest splits, bars and data biases — Units 1–2

First ask: are there labels, and is there an agent acting?

The three ways a machine can learn

Type	What the data has	Exam example
Supervised	a label for every example	50,000 loans, each repaid or defaulted
Unsupervised	no labels — find structure	customer groups · odd transactions
Reinforcement	an agent, actions and rewards	order-slicing bot, low-cost reward

Supervised splits in two

- Number → regression (next month's return)
- Category → classification (default yes/no)

Unsupervised does two jobs

- Find groups → clustering
- Find the odd ones → anomaly detection

Labels → supervised · no labels → unsupervised · rewards → reinforcement

Good on training data but bad on new data means overfitting

The only score that counts is the one on data the model has not seen

The gap gives it away (intern Mina, Q5)

train 99% *vs.* test 51% $\Rightarrow$ overfitting

Diagnose the fit

- Good train, bad test $\rightarrow$ overfitting
- Bad on both $\rightarrow$ underfitting
- 51% on two classes $\approx$ a coin flip

Cures for overfitting

- A simpler model or regularisation
- More data
- A held-out test set (next page)

Overfitting memorises noise; underfitting misses the pattern

Shuffle, then split: the standard test — time series is the special case

Q6–7 — hold out data the model has never seen

1 Standard — independent rows (e.g. 10,000 loan applications)

shuffle the rows at random, then hold out 20% as the test set



✓ train and test look alike — a fair score

2 Special case — a time series (daily data 2015–2024)

keep the time order: train on the past, test on the future



shuffling here would leak future days into training

 train  test

Independent rows → random split · time series → keep the time order

Dollar bars follow activity; biased data makes results look too good

How the data is sampled and what it leaves out

Dollar bars: one bar per fixed dollar amount (\$ millions)

$$\text{bars} \approx \frac{\text{dollars traded}}{\text{bar size}} : \quad \frac{400}{10} = 40 \quad \text{vs.} \quad \frac{100}{10} = 10$$

Bars

- Time bars close on the clock
- Dollar bars close per fixed \$ traded
- 4× the trading → about 4× the bars

Two biases

- Survivorship: dead funds left out
- Look-ahead: data used before release
- Cure: point-in-time data

Missing losers or future data make a backtest look better than it is

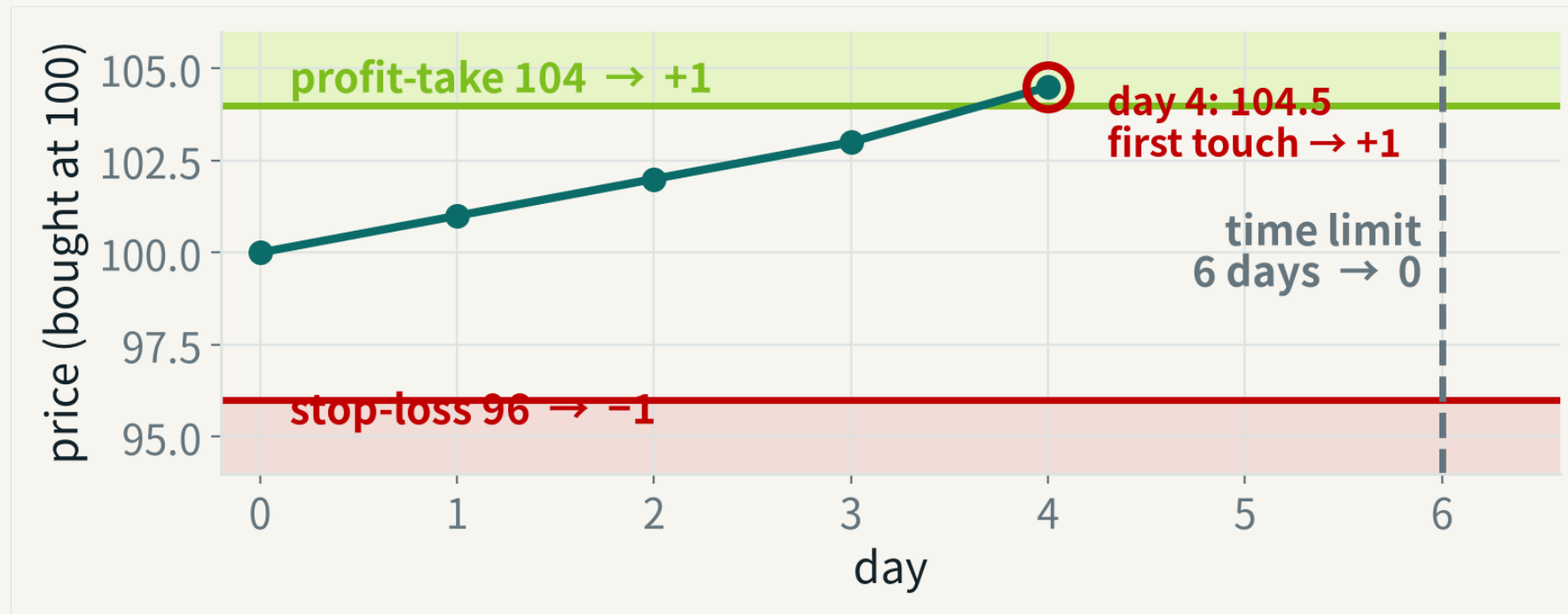
02

Labels and Linear Models

Triple-barrier labelling, least squares and regularisation — Units 3–4

The triple barrier is a race: the first barrier touched sets the label

Q12 — buy at 100, profit-take 104, stop-loss 96, time limit 6 days



Read the path in time order and stop at the first touch

Upper first is +1, lower first is -1, time limit first is 0

Triple-barrier labelling, Q12–14

The three outcomes

$y = +1$ (upper first), $y = -1$ (lower first), $y = 0$ (time limit)

How to label

- Set upper, lower and time barriers
- Walk the prices day by day
- Stop at the first barrier touched

The exam case

- 101, 102, 103, 104.5 → +1
- Falls to 96 before 104 → -1
- Stays in 96–104 for 6 days → 0

The label is set by which barrier comes first, not by the final price

OLS minimises squared errors; Lasso adds an L1 penalty

Fitting a line, then shrinking it

Ordinary least squares

$$\min \text{SSE} = (y_1 - \hat{y}_1)^2 + \dots + (y_n - \hat{y}_n)^2$$

Lasso

$$\text{SSE} + \lambda (|\beta_1| + \dots + |\beta_k|)$$

OLS

- Error = actual – fitted
- Squaring: signs count equally
- Big misses are punished more

Lasso (L1)

- Penalty on absolute values
- Sets weak coefficients exactly to 0
- 400 weak factors → keeps a few

Lasso = shrink and delete → feature selection

Ridge shrinks without deleting; Elastic Net does both

L2 and L1 + L2 penalties

Ridge

$$\text{SSE} + \lambda (\beta_1^2 + \dots + \beta_k^2)$$

Elastic Net

$$\text{SSE} + \lambda_1 \sum |\beta_j| + \lambda_2 \sum \beta_j^2$$

Ridge (L2)

- Penalty on squares
- Shrinks, never exactly to 0
- Keeps both of two correlated factors

Elastic Net (L1 + L2)

- Deletes noise like Lasso
- Shrinks groups like Ridge
- Match: Lasso L1 · Ridge L2 · EN both

Lasso deletes · Ridge shrinks · Elastic Net does both

03

Precision, Recall and F1

The confusion matrix and three stories: airport, COVID and card fraud

Recall reads across the actual row; precision reads down the predicted column

COVID story — 1,000 tested, 100 infected

	Predicted (test result)		
	positive	negative	
Actually infected	TP 90	FN 10	Recall = across this row $90 / 100 = 0.90$
Not infected	FP 45	TN 855	

↓

Precision = down this column: $90 / 135 \approx 0.67$

FN = sick, sent home
FP = healthy, extra PCR

Second word = the prediction · first word = was it right?

Recall asks how many we caught; precision asks how many flags were right

Card fraud — 20 frauds in 10,000 transactions; 50 blocked, 15 of them fraud

Definitions

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}$$

Card fraud (Q30–31)

$$R = \frac{15}{20} = 75\%, \quad P = \frac{15}{50} = 30\%$$

Pick by cost

- Missed case costly (FN) → recall
- False alarm costly (FP) → precision
- COVID: cheap PCR re-test → recall

Read the question

- 'How many frauds stopped?' → recall
- 'How many blocks right?' → precision
- False-alarm complaints → precision

FN hurts → maximise recall · FP hurts → maximise precision

When the target is rare, accuracy can be high and useless

Airport story — 10 terrorists among 1,000,000 passengers

Model A says 'civilian' to everyone

$$\text{Accuracy} = \frac{TP + TN}{N} : \frac{0 + 999,990}{1,000,000} = 99.999\%$$

Two lazy models

- A, all civilian: 99.999% accurate, R = 0%
- B, all terrorist: recall 100%
- B's precision: $10 / 1,000,000 = 0.001\%$

Lesson

- Rare target → accuracy misleads
- Recall alone is gamed by flagging all
- Precision stops that trick: need both

Choose the KPI that matches the mission, not the one that looks best

F1 is the harmonic mean, high only when precision and recall are both high

One score for two competing goals

F1 score, worked on Model C (Q24)

$$F_1 = \frac{2PR}{P+R} : \quad P=0.4, \quad R=0.8 \Rightarrow F_1 = \frac{2(0.4)(0.8)}{1.2} \approx 0.53 < 0.60$$

Why harmonic

- Pulled toward the smaller value
- $P=1, R=0 \rightarrow F1=0$ (mean 0.5)
- So below the simple average

Pick the model

- A: $F1=0$
- B: $F1 \approx 0$
- C: $F1 \approx 0.53 \rightarrow$ choose C

Compute $2PR / (P + R)$, not $(P + R) / 2$

04

Classifiers and Anomaly Detection

Logistic regression, SVM, LOF, One-Class SVM and Isolation Forest — Units 5–6

Logistic regression gives a probability for a yes/no outcome

The sigmoid squashes any score into 0–1

Logistic regression

$$p = \sigma(z) = \frac{1}{1 + e^{-z}}, \quad z = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k$$

Read it

- z can be any number
- $\sigma(z)$ is always between 0 and 1
- Above the cut-off (often 0.5) → yes

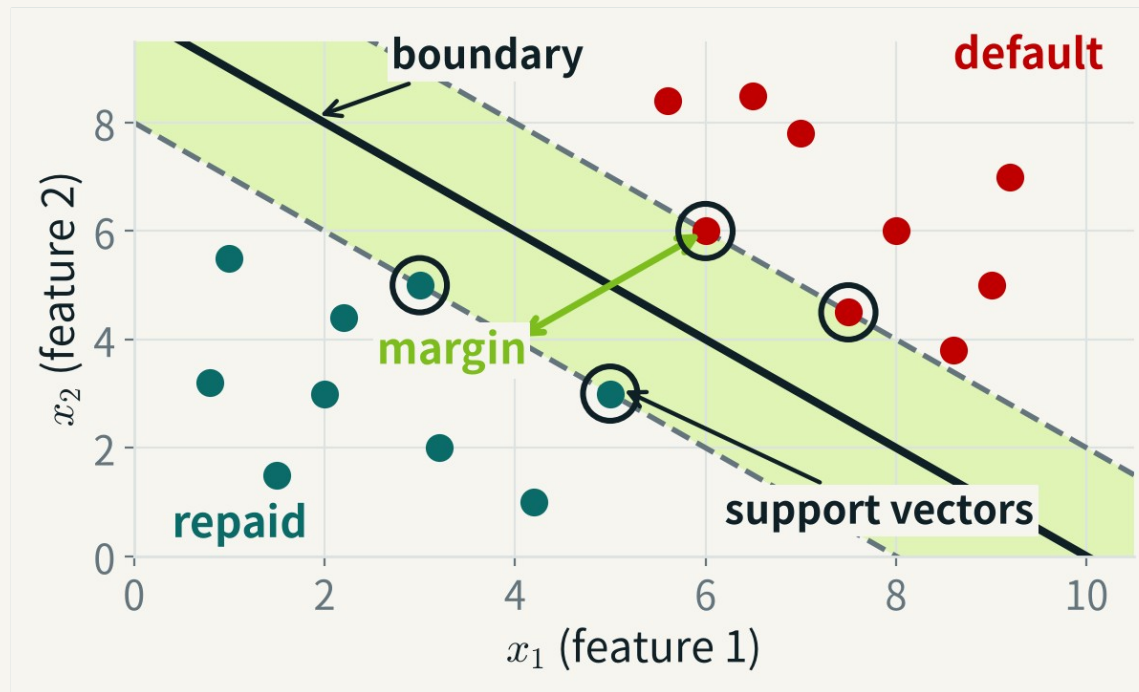
Use it when

- The outcome is yes/no (default?)
- You need a probability, not a label
- Signs show each input's direction

Need a probability for a yes/no outcome → logistic regression

An SVM puts the boundary in the middle of the widest gap

The usual picture: two features in the x-y plane, as in linear regression



Read the graph

- Dot = one customer
- Solid line = the boundary
- Dashed lines = margin edges
- Widest margin wins
- Circled = support vectors
- Exam Q34–35, one axis: 2, 3, 4 vs 8, 9 → boundary at 6

Only the closest points — the support vectors — fix the boundary

Support vectors alone decide where the SVM boundary sits

Maximum margin, worked on one axis

The exam case

$$\text{boundary} = \frac{4+8}{2} = 6, \quad \text{margin} = 2 \text{ on each side}$$

Margin

- Find the closest point of each class
- Gap runs 4 → 8
- Boundary at its middle: 6

Support vectors

- The closest points: 4 and 8
- Moving 2, 3 or 9 changes nothing
- Moving 4 or 8 moves the boundary

SVM boundary = middle of the widest empty road

Unsupervised detectors find the odd ones without fraud labels

LOF and One-Class SVM (Isolation Forest comes after the trees)

LOF — Local Outlier Factor

$$\text{LOF}(x) = \frac{\text{density of its neighbours}}{\text{density of } x} \gg 1 \Rightarrow \text{anomaly}$$

The ideas

- LOF: much lonelier than its neighbours
- One-Class SVM: learns what normal is
- Neither needs fraud labels

When to use

- Only normal data → One-Class SVM
- Local density differs → LOF
- No labels at all → any unsupervised one

No labels and 'find the odd ones' → unsupervised anomaly detection

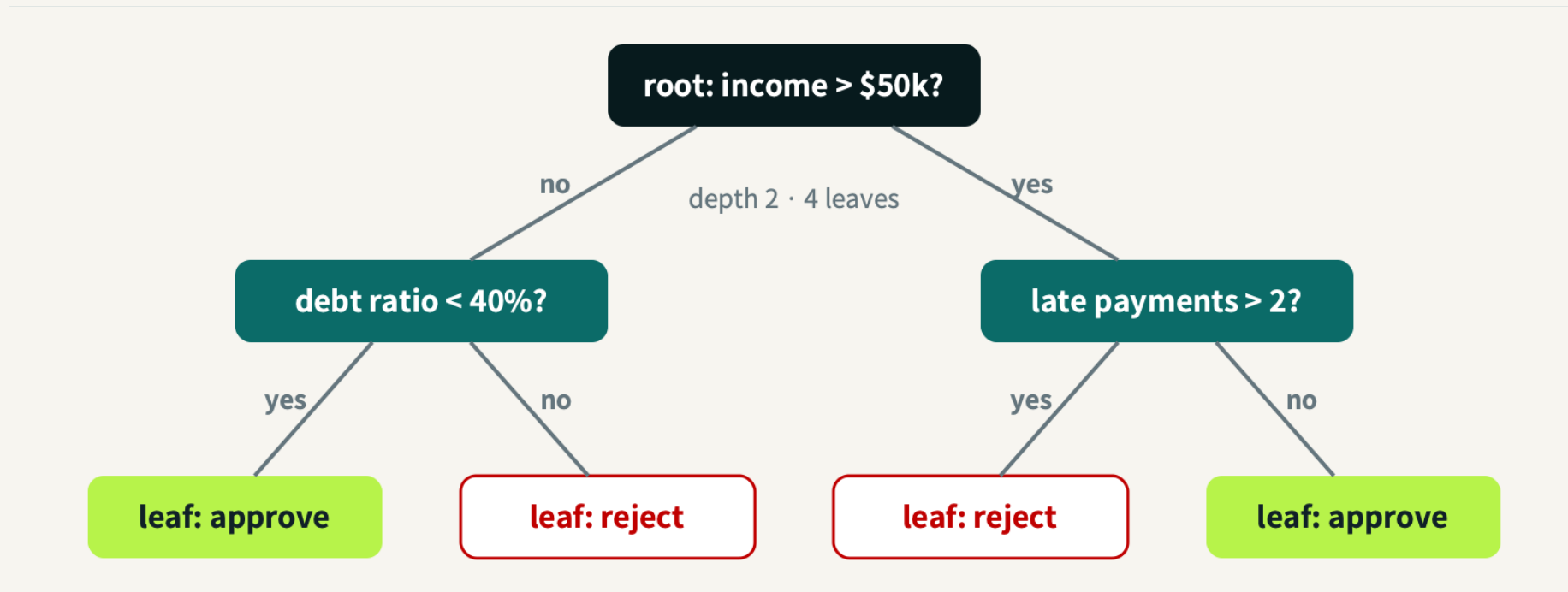
05

Trees and Ensembles

Decision trees, forests, bagging, boosting and Isolation Forest — Unit 7

A decision tree asks yes/no questions until it reaches a leaf

Credit story — a tree built from 2,000 past loans



Root question → branches → leaf holding the decision

A tree picks the purest split; grown to the end, it memorises

Two ways to measure impurity: Gini and entropy

Impurity of a node (lower = purer; p = share of each class)

$$\text{Gini} = 1 - \sum_k p_k^2$$

$$\text{Entropy} = -\sum_k p_k \log_2 p_k$$

How it decides

- Pick the split that lowers impurity most
- Pure node: Gini = 0, entropy = 0
- 50/50 node: Gini 0.5, entropy 1 bit

Overfitting

- Fully grown tree: near 100% on train
- Poor on new applicants
- Stop it early: the four controls

A fully grown tree overfits — limit how far it grows

Four controls keep a tree small

Q39–42 — all four are defences against overfitting

Control (exam value)	What it means	Simpler tree when
max_depth = 4	at most 4 questions from root to any leaf	smaller
max_leaf_nodes = 8	at most 8 leaves (final decisions)	smaller
min_samples_split = 50	split a node only if it holds ≥ 50	larger
min_samples_leaf = 30	every leaf must hold ≥ 30	larger

Max controls: go smaller · min controls: go larger

A random forest votes for a class and averages for a number

500 trees, each on a random sample of the loans

Classification (Q43)

class : majority vote, $\frac{320}{500} = 0.64 \approx P(\text{default})$

Regression (Q44)

number : $\hat{y} = (\hat{y}_1 + \hat{y}_2 + \dots + \hat{y}_{500}) / 500$

Why a forest

- Each tree sees a bootstrap sample
- Random features at each split
- Errors cancel → lower variance

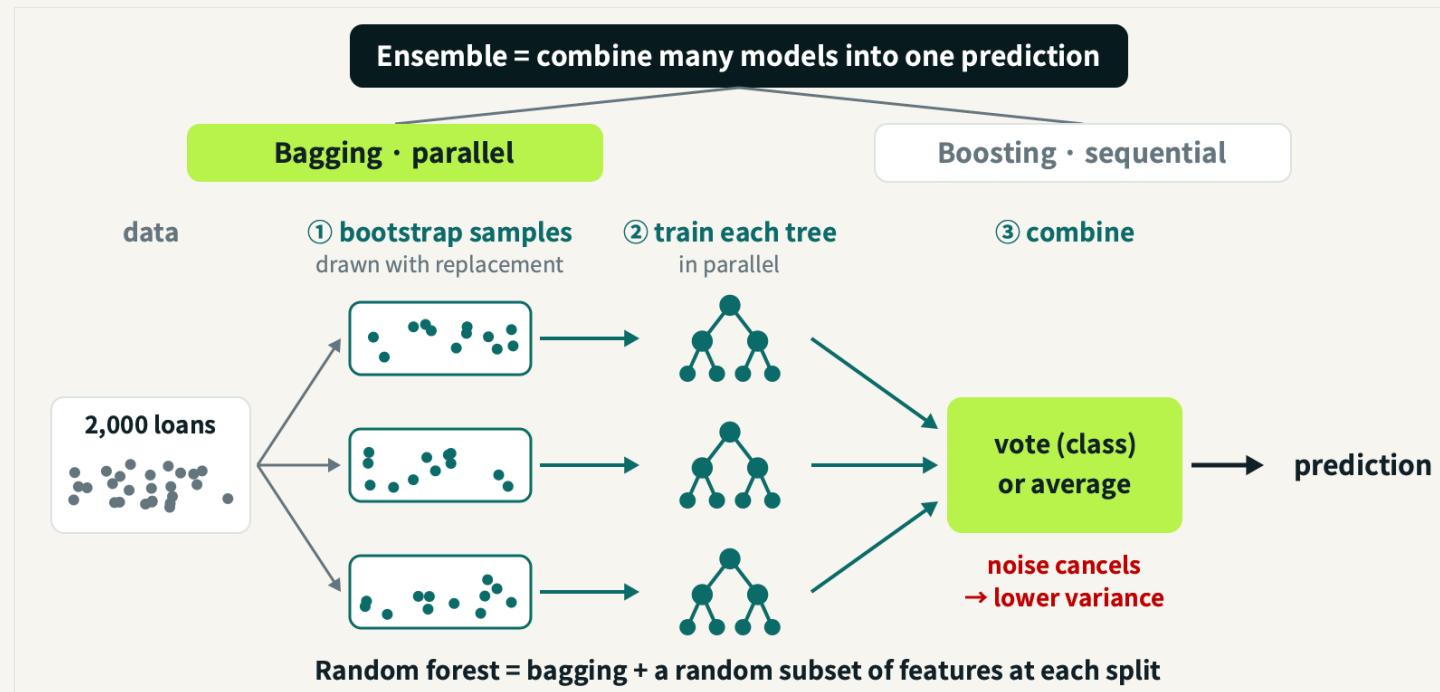
Combine

- Class → majority vote (320 of 500)
- Number → average of 500 trees
- Vote share $\approx$ probability

Classification → vote • regression → average

Bagging trains many trees side by side and combines them

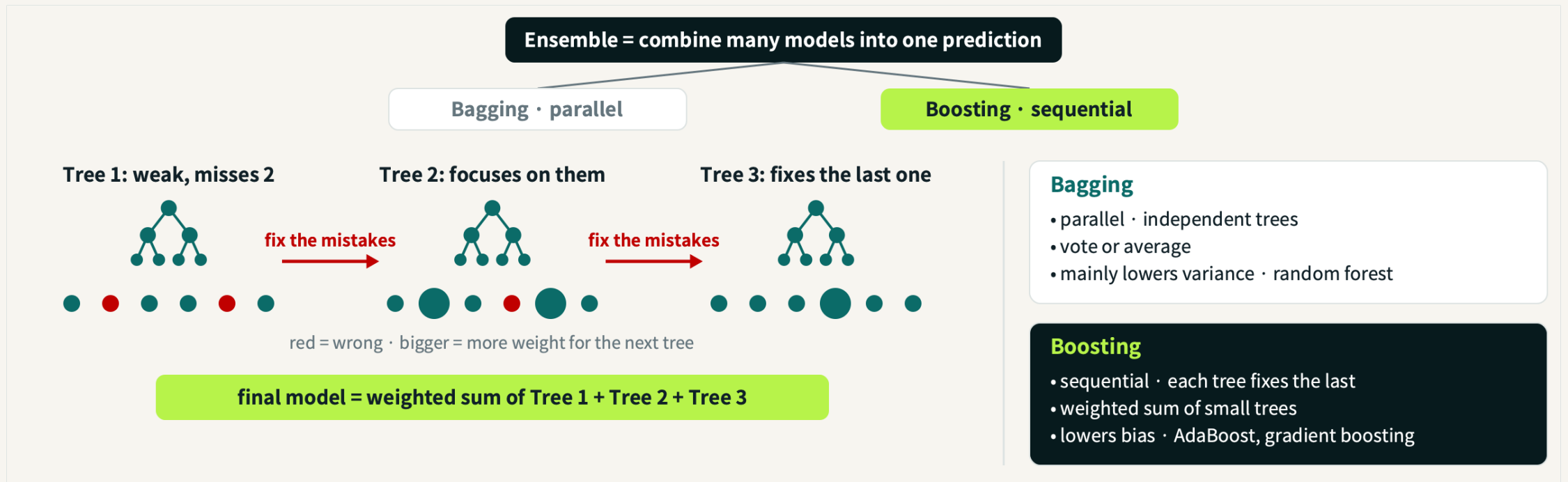
Q45 — bootstrap aggregating, one of the two ensemble families; a random forest is bagging



Bagging = parallel trees on bootstrap samples → vote or average

Boosting trains small trees one after another, each fixing the last

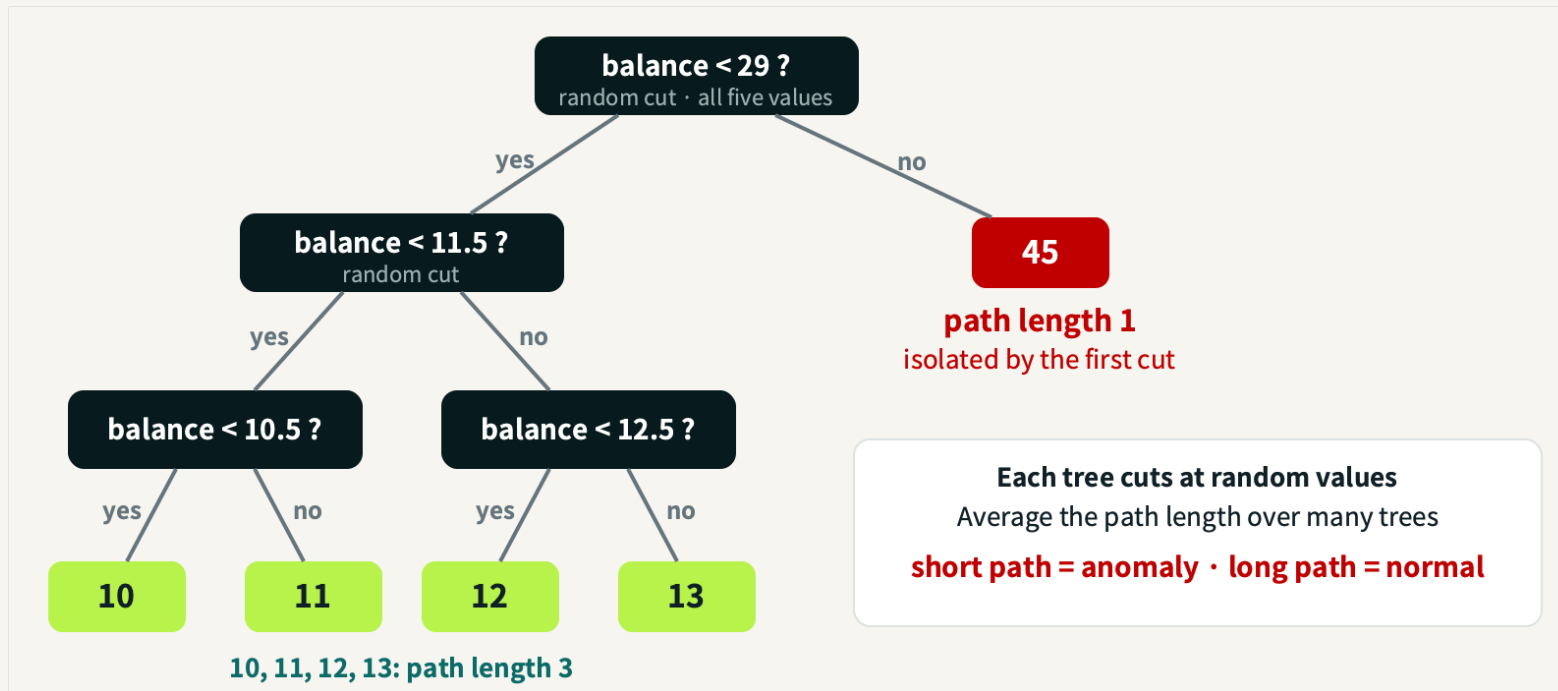
Q46–47 — the other ensemble family, and how the two compare



Bagging: parallel, lowers variance · boosting: sequential, lowers bias

Isolation Forest uses random trees to isolate an anomaly in a few cuts

Q48 — trees for anomaly detection: balances 10, 11, 12, 13 and 45 (thousand dollars)



Short path to isolation = anomaly

Ten triggers worth memorising (1/2)

If the question shows this, think that

If you see...

Labelled past examples

Think...

Supervised → regression or classification

99% on training, 51% on new data

Overfitting

Making a test set

Shuffle, hold out 20% (time series: keep order)

Dead funds missing · future data used

Survivorship bias · look-ahead bias

Path hits a barrier

First touch decides: +1, -1 or 0

Check the data before the model

Ten triggers worth memorising (2/2)

If the question shows this, think that

If you see...

Many weak factors, keep only a few

Rare positives and 99.99% accuracy

Missed cases are dangerous

Yes/no outcome needing a probability

Point alone in empty space

Think...

Lasso (L1) — sets coefficients to 0

Accuracy misleads → check recall

Recall — and F1 for two goals

Logistic regression

Anomaly (LOF · iForest · One-Class SVM)

Match the metric to the mission

Check the data first

Pick the metric that fits the mission

Then keep the model simple

**Know the rule behind
every answer on these pages**